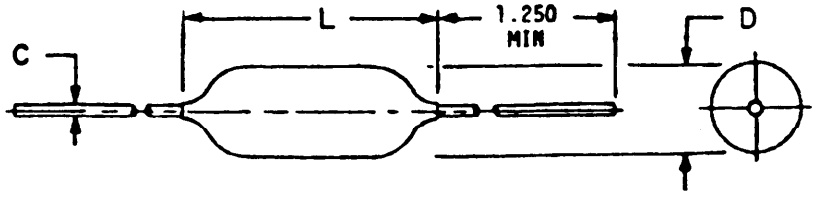
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY, STYLES CKR25, CKR26, AND CKR27

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39014.



Style	Dimensions		
	L	D	C
CKR25	.150±.020	.100 max	.020±.002
CKR26	.240±.020	.100 max	.020±.002
CKR27	.370±.030	.150 max	.025±.002

INCHES	MM
.002	0.05
.020	0.51
.025	0.64
.030	0.76
.100	2.54
.150	3.81
.240	6.10
.370	9.40
1.000	25.40
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead length may be a minimum of one inch long for use in tape and reel automatic insertion equipment, when specified in the ordering data.

FIGURE 1. Capacitors, styles CKR25, CKR26, CKR27 dimensions and configuration.

FSC 5910

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Tubular, conformal coated.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: See table I.

Radiographic inspection (failure rate level "S" (0.001 percent/1,000 hours) only): In accordance with MIL-C-39014.

Dissipation factor: Not more than 2.5 percent for BX; not more than 3.0 percent for BR.

Dielectric withstanding voltage: Body insulation shall be able to withstand 500 V dc between leads and case.

Insulation resistance:

At 25°C: Not less than 100,000 megohms or 1,000 megohm-microfarads whichever is less.

At 125°C: Not less than 10,000 megohms or 100 megohm-microfarads whichever is less.

Seal: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-39014.

Symbol	Capacitance change with reference to 25°C	
	Steps A through D of table V Bias = 0 volts	Steps E through G of table V Bias = rated voltage
BX	±15 percent	+15, -25 percent as applicable (see table I)
BR	±15 percent	+15, -40 percent as applicable (see table I)

Immersion: IR not less than 50 percent of initial 25°C requirement.

Salt spray (corrosion): Not applicable.

Resistance to soldering heat: ΔC +20, -0 percent from initial measured value.

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Life:

Rated conditions - 100 percent of dc rated voltage at 125°C.

0 through 3,000 hours:

IR - At 25°C, not less than 50 percent of initial 25°C requirement.

At 125°C, not less than 50 percent of initial 125°C requirement.

ΔC - ±20 percent from initial measured value.

DF - Not more than 2.5 percent for BX; not more than 3.0 percent for BR.

4,000 through 32,000 hours:

IR - At 25°C, not less than 15 percent of initial 25°C requirement.

At 125°C, not less than 15 percent of initial 125°C requirement.

ΔC - ±20 percent from initial measured value.

DF - Not greater than 2.5 percent.

Accelerated condition - 200 percent of dc rated voltage at 125°C.

0 and 250 hours:

IR - At 25°C, not less than 50 percent of initial 25°C requirement.

At 125°C, not less than 50 percent of initial 125°C requirement.

ΔC - ±20 percent from initial measured value.

DF - Not more than 2.5 percent.

1,000, 2,000 and 4,000 hours:

IR - At 25°C, not less than 15 percent of initial 25°C requirement.

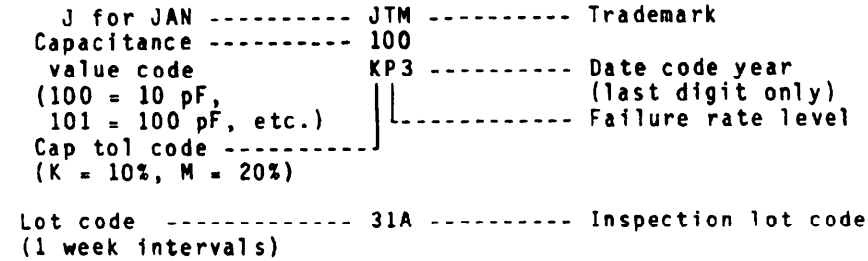
At 125°C, not less than 15 percent of initial 125°C requirement.

ΔC - ±20 percent from initial measured value.

DF - Not more than 2.5 percent.

Part number: M39014/25-(dash number from table I).

Marking: In accordance with MIL-C-39014 except that minimum marking as shown in the following example will be permitted for styles CKR25 and CKR26. Full marking shall be included on the package.



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TABLE I. Dash number and electrical characteristics.

Dash number				Capacitance	Capacitance tolerance	DC rated voltage
Failure rate level (%/1,000 hours)						
1.0 (M)	0.1 (P)	0.01 (R)	0.001 (S)			
Style CKR25, voltage-temperature limits of ±15% and +15, -25%						
				pF	percent	volts
0001	0201	0401	0601	10	10	100
0002	0202	0402	0602	10	20	"
0003	0203	0403	0603	12	10	"
0004	0204	0404	0604	15	10	"
0005	0205	0405	0605	15	20	"
0006	0206	0406	0606	18	10	"
0007	0207	0407	0607	22	10	"
0008	0208	0408	0608	22	20	"
0009	0209	0409	0609	27	10	"
0010	0210	0410	0610	33	10	"
0011	0211	0411	0611	33	20	"
0012	0212	0412	0612	39	10	"
0013	0213	0413	0613	47	10	"
0014	0214	0414	0614	47	20	"
0015	0215	0415	0615	56	10	"
0016	0216	0416	0616	68	10	"
0017	0217	0417	0617	68	20	"
0018	0218	0418	0618	82	10	"
0019	0219	0419	0619	100	10	"
0020	0220	0420	0620	100	20	"
0021	0221	0421	0621	120	10	"
0022	0222	0422	0622	150	10	"
0023	0223	0423	0623	150	20	"
0024	0224	0424	0624	180	10	"
0025	0225	0425	0625	220	10	"
0026	0226	0426	0626	220	20	"
0027	0227	0427	0627	270	10	"
0028	0228	0428	0628	330	10	"
0029	0229	0429	0629	330	20	"
0030	0230	0430	0630	390	10	"
0031	0231	0431	0631	470	10	"
0032	0232	0432	0632	470	20	"
0033	0233	0433	0633	560	10	"
0034	0234	0434	0634	680	10	"
0035	0235	0435	0635	680	20	"
0036	0236	0436	0636	820	10	"
0037	0237	0437	0637	1,000	10	"
0038	0238	0438	0638	1,000	20	"
0039	0239	0439	0639	1,200	10	"
0040	0240	0440	0640	1,500	10	"
0041	0241	0441	0641	1,500	20	"
0042	0242	0442	0642	1,800	10	"
0043	0243	0443	0643	2,200	10	"
0044	0244	0444	0644	2,200	20	"
0045	0245	0445	0645	2,700	10	"
0046	0246	0446	0646	3,300	10	"
0047	0247	0447	0647	3,300	20	"
0048	0248	0448	0648	3,900	10	"
0049	0249	0449	0649	4,700	10	"

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TABLE I. Dash number and electrical characteristics - Continued.

Dash number				Capacitance	Capacitance tolerance	DC rated voltage
Failure rate level (%/1,000 hours)						
1.0 (M)	0.1 (P)	0.01 (R)	0.001 (S)			
Style CKR25, voltage-temperature limits of ±15% and +15, -25%						
				pF	percent	volts
0050	0250	0450	0650	4,700	20	100
0051	0251	0451	0651	5,600	10	50
0052	0252	0452	0652	6,800	10	"
0053	0253	0453	0653	6,800	20	"
0054	0254	0454	0654	8,200	10	"
0055	0255	0455	0655	10,000	10	"
0056	0256	0456	0656	10,000	20	"
Style CKR26, voltage-temperature limits of ±15% and +15, -25%						
				pF	percent	volts
0057	0257	0457	0657	5,600	10	100
0058	0258	0458	0658	6,800	10	"
0059	0259	0459	0659	6,800	20	"
0060	0260	0460	0660	8,200	10	"
0061	0261	0461	0661	10,000	10	"
0062	0262	0462	0662	10,000	20	"
0063	0263	0463	0663	12,000	10	50
0064	0264	0464	0664	15,000	10	"
0065	0265	0465	0665	15,000	20	"
0066	0266	0466	0666	18,000	10	"
0067	0267	0467	0667	22,000	10	"
0068	0268	0468	0668	22,000	20	"
0069	0269	0469	0669	27,000	10	"
0070	0270	0470	0670	33,000	10	"
0071	0271	0471	0671	33,000	20	"
0072	0272	0472	0672	39,000	10	"
0073	0273	0473	0673	47,000	10	"
0074	0274	0474	0674	47,000	20	"
Style CKR27, voltage-temperature limits of ±15% and +15, -25%						
				pF	percent	volts
0075	0275	0475	0675	12,000	10	100
0076	0276	0476	0676	15,000	10	"
0077	0277	0477	0677	15,000	20	"
0078	0278	0478	0678	18,000	10	"
0079	0279	0479	0679	22,000	10	"
0080	0280	0480	0680	22,000	20	"
0081	0281	0481	0681	27,000	10	"
0082	0282	0482	0682	33,000	10	"
0083	0283	0483	0683	33,000	20	"
0084	0284	0484	0684	39,000	10	"

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TABLE I. Dash number and electrical characteristics - Continued.

Dash number				Capacitance	Capacitance tolerance	DC rated voltage
Failure rate level (%/1,000 hours)						
1.0 (M)	0.1 (P)	0.01 (R)	0.001 (S)			
Style CKR27, voltage-temperature limits of±15% and +15, -25%						
				pF	percent	volts
0085	0285	0485	0685	47,000	10	100
0086	0286	0486	0686	47,000	20	100
0087	0287	0487	0687	56,000	10	50
0088	0288	0488	0688	68,000	10	"
0089	0289	0489	0689	68,000	20	"
0090	0290	0490	0690	82,000	10	"
0091	0291	0491	0691	100,000	10	"
0092	0292	0492	0692	100,000	20	"
Style CKR27, voltage-temperature limits of±15% and +15, -40%						
0093	0293	0493	0693	56,000	10	100
0094	0294	0494	0694	68,000	10	"
0095	0295	0495	0695	68,000	20	"
0096	0296	0496	0696	82,000	10	"
0097	0297	0497	0697	100,000	10	"
0098	0298	0498	0698	100,000	20	"
0099	0299	0499	0699	120,000	10	50
0100	0300	0500	0700	150,000	10	"
0101	0301	0501	0701	150,000	20	"
0102	0302	0502	0702	180,000	10	"
0103	0303	0503	0703	220,000	10	"
0104	0304	0504	0704	220,000	20	"
0105	0305	0505	0705	270,000	10	"

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AR, MI
Navy - EC
Air Force - 17, 85
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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